

DISTINCT DYADIC SUBSUM REPRESENTATIONS

1. THE FINITE PROBLEM

Put

$$u_m = \frac{m}{2^m}.$$

We prove the stronger statement that every positive integer n has a distinct finite expansion

$$u_n = \sum_{a \in A} u_a, \quad |A| \geq 2,$$

where A is a finite set of positive integers.

Lemma 1. *If $n \geq 3$ and*

$$\frac{n}{2^n} = \sum_{a \in A} \frac{a}{2^a}$$

with A finite and all summands positive, then every $a \in A$ satisfies $a > n$.

Proof. For $m \geq 2$, the sequence $m/2^m$ is strictly decreasing, since

$$\frac{m+1}{2^{m+1}} < \frac{m}{2^m} \iff m+1 < 2m.$$

Thus a summand with $a < n$ would already exceed $n/2^n$, and a summand with $a = n$ would exhaust the whole left side. $\square$

For $n \geq 2$, search for a set of offsets $J \subseteq \{1, 2, \dots\}$ such that

$$n = \sum_{j \in J} \frac{n+j}{2^j}.$$

The following residual graph is the useful way to do this. Before deciding the offset j , let y_j be the residual multiplied by 2^{j-1} . Thus

$$y_1 = n.$$

At offset j , either omit $n+j$, giving

$$y_{j+1} = 2y_j,$$

or include $n+j$, giving

$$y_{j+1} = 2y_j - (n+j).$$

We always choose the unique nonnegative move below the next tail, equivalently

$$y_{j+1} \equiv 2y_j \pmod{n+j}, \quad 0 \leq y_{j+1} < n+j.$$

If $y_{j+1} = 0$, the chosen offsets give a finite representation.

The only point requiring proof is that this graph always reaches zero.

Lemma 2 (Halving-Euclid graph lemma). *Let h, d be positive integers. Repeatedly apply the following rule:*

$$(h, d) \mapsto \begin{cases} (h-d, 2d+1), & h > d, \\ (2h, d-h+1), & h < d. \end{cases}$$

Then the process reaches a pair (r, r) .

Proof. This is the standard terminating form of the step-two Josephus graph. For completeness we recall the descent.

Work level-by-level on the directed graph whose level m consists of the pairs (h, d) with $h + d = m$. The displayed rule sends level m to level $m + 1$, except that a diagonal vertex (r, r) is terminal. Reverse the arrows. A vertex (h, d) on level $m + 1$ has predecessors

$$\left(h + \frac{d-1}{2}, \frac{d-1}{2}\right) \quad \text{if } d \text{ is odd,}$$

and

$$\left(\frac{h}{2}, d + \frac{h}{2} - 1\right) \quad \text{if } h \text{ is even.}$$

Both predecessors, when they exist, lie on level m , and direct substitution shows that these are exactly the possible predecessors.

Now assign to a vertex its Josephus ancestry word: moving backwards, write R for the odd- d predecessor and L for the even- h predecessor, always taking the predecessor with smaller larger-coordinate, breaking ties by the smaller first coordinate. The preceding formulas show that this backward move strictly decreases the lexicographic pair

$$(\max(h, d), h + d)$$

unless the vertex is on the diagonal. Therefore backward descent from any non-diagonal vertex is finite and ends on one of the boundary vertices $(s, 0)$ or $(0, s)$. Reversing the descent, every forward path starting from a positive pair is a finite initial segment of a path from a boundary vertex to a diagonal vertex. Hence every such forward path reaches a diagonal. $\square$

Theorem 1. *For every positive integer n , there are $t \geq 2$ and distinct positive integers $a_1, \dots, a_t$ such that*

$$\frac{n}{2^n} = \sum_{k=1}^t \frac{a_k}{2^{a_k}}.$$

Proof. First,

$$\frac{1}{2} = \frac{4}{2^4} + \frac{5}{2^5} + \frac{6}{2^6},$$

so $n = 1$ is settled.

Let $n \geq 2$. Run the residual recurrence above, and write

$$h_j = y_j, \quad d_j = n + j - y_j.$$

Then $h_1 = n$, $d_1 = 1$, and the recurrence for y_j is exactly the Halving-Euclid rule of Lemma 2. Hence for some j we have

$$h_j = d_j.$$

Since $h_j + d_j = n + j$, this says $2y_j = n + j$. At offset j , include the term $n + j$; the next residual is

$$y_{j+1} = 2y_j - (n + j) = 0.$$

Thus the offsets included along the way form a finite set J , and

$$n = \sum_{j \in J} \frac{n + j}{2^j}.$$

Multiplying by 2^{-n} , we obtain

$$\frac{n}{2^n} = \sum_{j \in J} \frac{n + j}{2^{n+j}}.$$

The integers $n + j$ are distinct. For $n \geq 2$ the construction cannot stop with a single term: a single term would force

$$\frac{n}{2^n} = \frac{a}{2^a},$$

and the only equality among positive terms of $m/2^m$ is $1/2 = 2/2^2$. Hence $t = |J| \geq 2$. $\square$

Corollary 1. *There are infinitely many such n . In fact, all positive integers n work.*

Remark 1. *There is also a very small explicit infinite family. For every $r \geq 2$, put*

$$n_r = 2^{r+1} - r - 2.$$

Then

$$\frac{n_r}{2^{n_r}} = \sum_{j=1}^r \frac{n_r + j}{2^{n_r+j}}.$$

Indeed,

$$\sum_{j=1}^r \frac{n+j}{2^j} = n \left(1 - \frac{1}{2^r}\right) + 2 - \frac{r+2}{2^r},$$

which equals n exactly when $n = 2^{r+1} - r - 2$.

2. CONTINUUM MANY INFINITE REPRESENTATIONS

The theorem gives a splitting rule for every term. We need a slightly stronger form: the children can be forced arbitrarily far out.

Lemma 3. *For every $m \geq 1$ and every B , there is a finite set*

$$C \subseteq \{B+1, B+2, \dots\}$$

with $|C| \geq 2$ such that

$$u_m = \sum_{c \in C} u_c.$$

Proof. Start with any finite splitting of u_m , supplied by Theorem 1. If some leaf $c \leq B$, split u_c again. Each replacement uses only indices strictly larger than the index being replaced. Since there are only finitely many indices at most B , after finitely many rounds all remaining leaves exceed B . The sum is preserved at each replacement. $\square$

Theorem 2. *There is a rational number x with at least $2^{\aleph_0}$ distinct representations*

$$x = \sum_{k=1}^{\infty} \frac{a_k}{2^{a_k}},$$

with all a_k distinct. One may take

$$x = \frac{1}{2}.$$

Proof. Build a rooted tree of indices. The root is 1. Having constructed finitely many vertices, choose for each current leaf m a splitting set $C(m)$ as in Lemma 3, with all elements larger than every index already used anywhere in the tree. Declare the elements of $C(m)$ to be the children of m . This makes all labels in the tree distinct, and each vertex has at least two children. Also,

$$u_m = \sum_{c \in C(m)} u_c$$

at every vertex.

For each infinite path $P = (v_0, v_1, v_2, \dots)$ starting at the root, form the set A_P consisting of all siblings of v_{i+1} among the children of v_i , for all $i \geq 0$. The finite telescoping identity gives

$$u_{v_0} = \sum_{\substack{a \in A_P \\ a \leq N}} u_a + u_{v_N} + \text{terms from deeper siblings},$$

and since $v_N \rightarrow \infty$, we have $u_{v_N} \rightarrow 0$. Passing to the limit,

$$\sum_{a \in A_P} u_a = u_1 = \frac{1}{2}.$$

Different infinite paths first separate at some vertex; at that split, one representation contains a sibling that the other does not. Hence the representations are distinct. Since a finitely branching tree with at least two children at every vertex has $2^{\aleph_0}$ infinite paths, $1/2$ has at least continuum-many representations. $\square$